

Clinical Research

Recurrence After a First Untreated Seizure in the Hong Kong Chinese Population

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Summary: *Purpose:* There is wide variation in the reported prognosis after a first unprovoked generalized tonic-clonic convulsion and in the risk factors that are associated with recurrence. Estimates for the risk of recurrence range from 26 to 71%. We investigated the likelihood of a second attack in Hong Kong Chinese patients.

Methods: One hundred thirty-two patients with a first convulsion that was unexplained by acute neurological or medical causes were retrospectively ascertained. Patients' demographic details, potential risk factors for recurrence, and current seizure status were recorded. Survival analysis was performed using the Kaplan-Meier procedure.

Results: The cumulative probability of a second attack at 1, 2, 3, and 4 years was 30, 37, 42, and 47%, respectively. Seizures in patients with abnormal computer tomography scans of the brain were associated with an increased risk of recurrence on multivariate analysis.

Conclusions: Thirty percent of the sample population experienced a second seizure after 1 year. An additional 17% continue to be at risk of a second convulsion during the next 3 years. **Key Words:** First seizure—Recurrence rate—Chinese population—Risk factors—Prognosis.

A first unexplained seizure is a common clinical problem; up to 5% of the population will experience an afebrile seizure during their lifetimes (1). In patients with a single convulsive episode, a major factor influencing the decision to start antiepileptic drugs is the likelihood of recurrence. However, there is uncertainty over this point because the reported risks of recurrent seizures vary from 26 to 71% (2–16). As a result, there is controversy regarding the management of this condition; U.S. neurologists are more likely to start therapy after one episode than are their U.K. counterparts (17,18). The earlier initiation of anticonvulsants may prevent further seizures and may reduce the long-term tendency to have seizures. These advantages should be balanced with the potential side effects from antiepileptic drugs as well as the social and psychological implications of labeling a person as an epileptic. In practice, many patients do not have a previous or concomitant neurological condition that would

help guide the clinician in determining the risks. Seizure type, a family history of epilepsy or of neurological disease or of prior febrile convulsion, abnormal neurological examination, Todd's paresis, and abnormal findings from EEG and neuroimaging have been implicated as predictors for recurrence (3,8,12). The seizure recurrence rate in the Chinese is uncertain; no studies that address this issue were found from a MEDLINE search of the period 1966–1999. This retrospective study aims to describe the relapse rate after a first convulsion and the factors associated with recurrence.

METHODS

We reviewed the records of 2938 referrals to the EEG laboratory from June 1993 to December 1996 at the Prince of Wales hospital in Hong Kong. Three hundred ten (11%) requests were for a first unprovoked seizure (19); the remaining 2628 EEGs were for other indications, such as psychiatric screening (35%), evaluation of patients with epilepsy on treatment (31%), acute medical emergencies such as central nervous system (CNS) infection or confusional state (15%), and miscellaneous conditions such as dementia and headache (19%). EEG

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and routine blood tests are performed as part of the recommended work-up of every person with a first convulsion in our institution. These investigations would not incur additional costs to the patient. Inclusion criteria consisted of the following. (a) Only those with a definite witnessed attack consisting of an unprovoked generalized tonic-clonic seizure were included. Partial seizures were not studied because they may be brief, the date of onset may not be reliably remembered by the patient, and there may be more diagnostic uncertainty. (b) Patients were older than 14 years. (c) Antiepileptic medications were withheld after the first seizure. We excluded (a) patients with questionable seizures such as confusion, abnormal behavior, unconsciousness, or amnesia; (b) patients with precipitant factors on admission such as cerebrovascular disease, trauma, CNS infection, or metabolic encephalopathy; (c) patients with other seizure types, such as myoclonus, absence, and status epilepticus; (d) patients with abnormal baseline investigations; only patients with normal glucose and renal and liver function tests were included; and (e) patients with radiological evidence of underlying pathology such as brain tumors, brain abscess, or acute stroke that was thought to be the cause of the seizure.

Standardized EEG findings consisted of a 30-min recording that included hyperventilation and intermittent photic stimulation at 1–24 Hz using bipolar longitudinal and transverse montages. EEGs were retrieved and read by one of the authors (C.F.H.), who had no access to the recurrence status of patients. Findings were classified into three groups: (a) focal or generalized epileptiform discharges such as sharp or spike waves, (b) localized nonepileptic activity such as focal slowing or reduction in voltage, and (c) normal. Clinical variables that may be associated with an increased risk were included, such as a history of remote symptomatic brain insult, a family history positive for seizures, the relation to the sleep/awake cycle, an abnormal physical examination, and an abnormal CT brain scan.

Relevant information such as a family history of epilepsy or whether a second seizure had occurred was derived from medical and computer records and from telephone interviews. Recurrence rates were determined by survival analysis, taking an event as the occurrence of a second seizure. Survival analysis was performed using the Kaplan-Meier procedure with the results displayed as Kaplan-Meier curves: the cumulative probability of seizure recurrence was plotted as a function of time from the first convulsion. Univariate analyses with statistical significance level and 95% confidence interval of the crude relative risk were performed by the Cox proportional hazard model. This was followed by multivariate analysis using the Cox regression model of those statistically significant variables associated with recurrence in the univariate analysis.

RESULTS

We retrieved 310 records, of which 132 fulfilled the criteria as given here. The male-to-female ratio was 1:1, and age ranged from 13 to 86 years, with a mean of 33 years. Four patients had died (two from pneumonia, one from a motor vehicle accident, and one from an unknown cause), and four were lost to follow-up. Two patients had a second seizure before the EEG were performed and were included in the analysis. The mean interval between a patient's seizure and undergoing an EEG was 15.7 days (median 15.0 days, range 1–36 days). EEGs were normal in 73 patients (55%) and abnormal in 59 (45%). Mean duration of follow-up was 27.2 months, with a median of 27.7 months. We followed 87, 77, 49, and 21 patients for more than 1, 2, 3, and 4 years, respectively. Brain CTs were performed in 85 (64%) of patients. Neuroimaging abnormalities were found in nine patients (6.8%) and consisted of cerebral atrophy, arachnoid cysts, and evidence of subclinical cerebrovascular disease that were not predicted on the basis of the clinical history and examination. Those without CT scans were included in the analysis except for the univariate analysis of abnormal CT imaging as a risk factor; in which case this was only performed for patients who had available scans. The risk of recurrence at 1, 2, 3, and 4 years was 30, 37, 42, and 47%, respectively (Fig. 1). The characteristics of the patients and the results of univariate analysis are shown in Table 1. Those with seizures during sleep and with an abnormal CT brain scan were at an increased risk of having a second seizure on univariate analysis. However, after multivariate analysis, only abnormal CT brain scan was a predictor in the Cox regression model (relative risk ratio 2.44, 95% CI 1.09–5.44, $p < 0.03$); the other potential risk factor, nocturnal seizures, was not associated with a statistically significant higher risk (relative risk 0.61, 95% CI 0.3–1.25, $p = 0.18$).

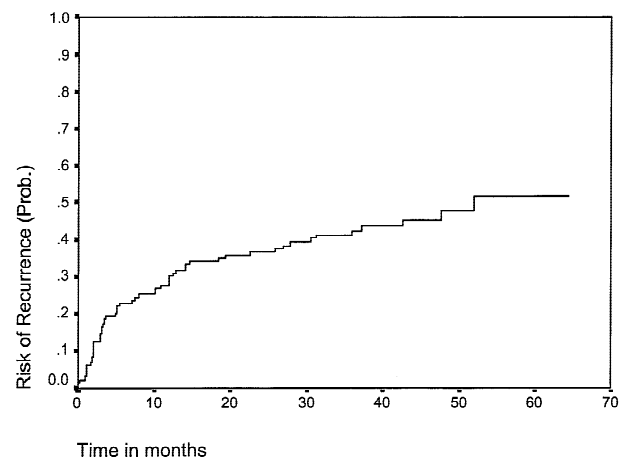


FIG. 1. Risk of recurrence after first unprovoked seizure in Chinese adults: Kaplan-Meier curve.

TABLE 1. Risk factors for recurrence: univariate analysis using Cox proportional hazards model

Risk factor	n (%)	Number of patients with recurrent seizures (%)	Rate ratio	95% confidence interval	p
Age (yr)	132 (100)	60 (47)	0.99	0.98–1.01	0.26
Male	66 (50)	27 (43)	1	1	—
Female	66 (50)	29 (45)	1.16	0.68–1.98	0.59
Symptomatic	13 (9.8)	7 (54)	1.57	0.71–3.49	0.27
Positive family history	3 (2.3)	2 (66)	2.06	0.50–8.48	0.32
Seizures awake	106 (80.3)	39 (37)	0.45	0.25–0.80	<0.01
Abnormal examination	7 (5.3)	5 (71)	2.18	0.86–5.52	0.10
Epileptiform EEG	39 (29.5)	20 (51)	1.39	0.73–2.65	0.31
Slow activity EEG	17 (12.9)	6 (35)	1.04	0.41–2.61	0.94
Abnormal CT brain	9 (6.8)	8 (89)	2.11	1.49–2.99	<0.01

DISCUSSION

In patients with epilepsy, the prognosis and treatment are best determined by establishing the epilepsy syndrome or the underlying cause for the seizures (20–22). Such a diagnostic label may not be identified in all cases at or after the first seizure, and the seizure remains unexplained. The variation found in recurrence rates in those who have experienced a first unprovoked seizure seems to be due to methodological differences or in the characteristics of the study populations. In the hospital-based study by Hauser et al. (8) of the risk of recurrence, the probability of a second attack was estimated to be 10, 24, and 29% at 1, 3, and 5 years, respectively, in those without previous CNS insult and 26, 41, and 48%, respectively, in those with a record of such insult (8). The National General Practice Survey of Epilepsy, a prospective population-based cohort study, found that 40% of patients had a second seizure 1 year after the initial attack (10). Other studies, such as that conducted by Elwes et al. (15), produced a much higher relapse rate, but the sample population was heterogeneous with different causes and follow-up was poor. If patients with seizure recurrence within 24 h were excluded, the recurrence rate in this study would have been lower (6). In a meta-analysis of 16 reports on the prognosis of first seizure by Berg and Shinnar (7), the three methodological factors that most influenced these variation were study selection criteria, retrospective or prospective ascertainment of patients, and interval between the first attack and the time at which risk was assessed. Even in studies with similar methodology, variation in prognosis can be due to the fact that the distribution of risk factors may differ and so produce different recurrence rates.

This study has several limitations. Retrospective ascertainment results in a higher estimate of recurrence due to follow-up bias: patients who experience a recurrence are more likely to be under the care of a physician and those who remain well may be lost. However, Berg and Shinnar (7) found that in studies with good follow-up,

this may not be as significant, and they commented on the good agreement between retrospective and prospective studies in the risk of recurrence in patients with idiopathic first convulsion and normal EEGs. In our study, patients were included only after the EEG, so the true recurrence rate might be underestimated; the patient may have a second attack by the time of diagnosis. We identified only two such cases. A prospective study would produce more accurate clinical information, more reliable case ascertainment, and uniform investigations. We attempted to overcome the problem with gaps in the initial assessment during the telephone follow-up: for example, we asked about the timing of the seizure or for a family history of epilepsy if this information had been omitted at the first consultation.

Although this study was conducted in only one center, our hospital was the only publicly funded acute hospital in the district during the study period. In Hong Kong, the majority of emergency and inpatient care is provided for by government hospitals; private institutions account for only 4% of all epilepsy- and convulsion-related admissions (23). In this study, neuroimaging abnormalities (consisting of cerebral atrophy, arachnoid cysts, and evidence of cerebrovascular disease) and a history of the first seizure occurring in sleep were associated with recurrence on univariate analysis, but multivariate analysis showed that only an abnormal CT brain scan was a predictor. Other previously identified risk factors were not found to be associated with a higher risk of recurrence: earlier studies have implicated the cause of seizures and EEG changes as major prognostic factors (7,2). The failure to identify predictors in this study may be limited by the sample size and the small number of patients with potential risk factors. Another possible reason is that patients with epileptiform EEGs would have been treated and would not therefore have been recruited: we found that among the patients who were not included into the study, antiepileptic drug treatment was not among the reasons for exclusion. EEGs performed earlier may

have provided more information and detected more abnormalities if performed within 24 h of the first attack (25).

This is the first report to address the likelihood of recurrence of a first seizure and the potential risk factors for a second attack in a Chinese population. In our patients, 37% had a second unprovoked seizure after 2 years and an additional 10% had a second episode in the next 2 years. This compares with an estimate of 42% from previous studies (7); differences from existing reports may be explained by intrinsic differences in the population studied, in methodology, or in the risk factor profile. The pattern of recurrence is similar to previous studies with the greatest risk (30%) within the first year and subsequent tapering in the years after this.

Because three fourths of those with two or three unprovoked seizures have further attacks, the suppression of further seizures by the start of antiepileptic drugs appears justified in this situation (26). Although there is some evidence that the start of treatment after a single seizure reduces the probability of a second attack (27), this is not a universal practice. One approach would be to lower the threshold for initiation of treatment in patients with factors that are shown to be highly predictive for relapse. Further prospective studies are required to determine risk factors that can guide the management of patients presenting with a first tonic-clonic seizure.

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